

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

**AN ONLINE DIGITAL ORDERING SYSTEM WITH MEMBERSHIP
MANAGEMENT**

A Capstone Project Proposal

In Partial Fulfillment for the Requirements in

Capstone 1 and Research Subject

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CHAPTER 1 INTRODUCTION

1.1 Background of the Study

The rapid advancement of digital technology in recent years has fundamentally altered how businesses operate and interact with their customers. The proliferation of the internet, cloud computing, and mobile applications has enabled businesses of all sizes to transition from manual, paper-based processes to fully automated digital systems. Among the most transformative developments in this era is the rise of online ordering platforms, which allow customers to browse products and services, place orders, and complete transactions without the need for in-person interaction. This shift has redefined the relationship between businesses and their customers, creating new expectations for convenience, speed, and transparency.

In the Philippine context, the digital revolution has arrived with particular momentum. According to the Department of Information and Communications Technology (DICT), the Philippine government has made digital transformation a cornerstone of its national development agenda through the Philippine Digital Transformation Strategy 2022. This strategy aims to leverage information and communications technology (ICT) to strengthen the economy, improve public services, and enhance the quality of life of Filipinos, with a special emphasis on enabling small and medium enterprises (SMEs) to adopt digital tools. The DICT's mandate extends to approximately 2 million SMEs and over 115 million citizens across 82 provinces, underscoring the scale and urgency of this national initiative.

Within this broader digital landscape, direct-selling businesses occupy a particularly significant niche in the Philippine economy. Data from Agriculture and Agri-Food Canada indicate that the retail value of direct selling in the Philippines reached approximately 1.37

billion US dollars in 2021, with projections forecasting growth to approximately 1.75 billion US dollars by 2026. The direct-selling industry in the Philippines is supported by a robust distributor network, often organized through tiered membership structures that incentivize performance and loyalty. Despite its economic significance, many direct-selling companies continue to rely on manual or semi-automated processes for order management, inventory control, and distributor communications, which limits their operational efficiency and growth potential.

Amazing World Marketing Corporation (AWMC) is a direct-selling company based in the Philippines that offers a diverse range of products, including wellness items, personal care products, health supplements such as Immukira-AG and Immuvit-ZinC, and the Ardeur de France fragrance line. The company features a tiered membership structure ranging from Silver to Diamond, with each level providing members with exclusive benefits, business packages, and pricing incentives. Like many similar enterprises, AWMC currently relies on manual order processing, which introduces a range of operational challenges including errors in order recording, delays in fulfillment, inventory discrepancies, and limited visibility into sales performance data.

Traditional ordering systems, which remain prevalent among small and medium-sized enterprises, depend on manual procedures such as verbal communication, handwritten order forms, and cash-based payment methods. These approaches are inherently prone to human error, inefficiency, and delay, which can result in customer dissatisfaction and lost business opportunities. Additionally, the absence of systematic data collection and analysis makes it difficult for business owners and managers to track sales trends, monitor customer behavior, and generate accurate reports. These limitations are compounded in membership-based

businesses like AWMC, where the need to track distributor activity, manage tier transitions, and administer loyalty rewards adds a further layer of complexity to daily operations.

To address these challenges, businesses across industries have increasingly adopted online digital ordering systems that automate order processing, reduce human error, and accelerate transaction speed. These systems provide customers with the convenience of ordering anytime and from any location, while simultaneously giving businesses greater control over their operations and data. When combined with membership management capabilities, online ordering platforms offer a comprehensive solution that enhances customer engagement, supports distributor retention, and enables data-driven decision-making. This convergence of ordering and membership management functions represents a significant opportunity for companies like AWMC to modernize their operations and compete more effectively in the digital economy.

In response to these identified needs, the research team proposes the development of the AWMC Ordering System, a web-based platform built using PHP, MySQL, HTML, CSS, and JavaScript. The proposed system is designed to automate and streamline AWMC's core operational functions, including product browsing, order placement, payment processing, inventory management, membership administration, and sales analytics. It will support multiple payment methods including GCash, PayMaya, PayPal, cash on pickup, and cash on delivery, catering to the diverse payment preferences of AWMC's customer base. Dedicated interfaces for administrators and customers will ensure that each user group has access to the tools and information relevant to their role, while an integrated analytics dashboard will provide management with real-time insights to inform strategic decisions.

The development of this system represents not only a practical solution to AWMC's operational challenges but also an applied demonstration of how web development, database management, and systems design principles can be leveraged to drive digital transformation in the Philippine direct-selling industry. By operationalizing the concepts derived from both international literature and local research, the AWMC Ordering System aims to deliver measurable improvements in efficiency, accuracy, and customer satisfaction, while supporting the sustainable growth of the company and its network of distributors.

1.2 Statement of the Problem

AWMC is currently facing a range of operational challenges that stem from its reliance on manual processes and the absence of a comprehensive, integrated digital ordering and management system. These inefficiencies affect not only the day-to-day operations of the business but also its ability to retain customers, support its distributor network, and compete in an increasingly digital marketplace. The lack of a centralized platform for order management, inventory tracking, membership administration, and sales reporting has created compounding difficulties across multiple aspects of AWMC's operations.

The primary issues confronting AWMC can be articulated through the following specific problem statements:

First, AWMC lacks a centralized platform for managing and tracking customer orders. Orders are currently processed through informal communication channels, including verbal agreements and handwritten records, which significantly increases the likelihood of human error. These errors may include incorrect order information, duplicate transactions, missed

orders, and inaccurate pricing. Without a systematic tracking mechanism, it is also difficult to monitor the real-time status of each order, resulting in delays in processing and fulfillment that frustrate both customers and employees. This absence of transparency and accountability creates confusion and inefficiency in the company's day-to-day operations.

Second, the company experiences significant difficulties in inventory management due to the absence of an automated system. Stock levels are updated manually, which leads to frequent inaccuracies. This can result in situations where items appear to be available in records but are actually out of stock, or conversely, where available stock is not captured in inventory records. The lack of real-time inventory monitoring makes it challenging for management to identify low-stock items in a timely manner, creating the risk of missed sales opportunities and customer dissatisfaction. Accurate and real-time inventory data are essential for maintaining product availability and supporting effective procurement planning.

Third, AWMC currently lacks a structured membership management system, which limits its ability to foster customer loyalty and retain its distributor network. Without a formal system for tracking member activity, managing tier transitions, and administering loyalty rewards, the company cannot efficiently deliver the personalized benefits and incentives that define its membership program. This limitation impairs AWMC's ability to build long-term relationships with its clients and distributors, and reduces the effectiveness of its loyalty and retention strategies.

Fourth, AWMC provides limited payment options, which restricts customer convenience and may deter potential buyers from completing transactions. In today's digital economy, customers expect a variety of payment methods, including digital wallets and online payment platforms. The absence of such options may discourage prospective customers,

particularly those who prefer cashless transactions, from patronizing the company. Offering diverse and flexible payment methods is essential for improving the customer experience and maximizing transaction completion rates.

Fifth, the company lacks robust real-time sales tracking and reporting capabilities. Without an automated system, sales data must be recorded and analyzed manually, which slows the decision-making process and introduces opportunities for error. Managers do not have timely access to accurate information on revenue, order trends, or product performance. This limitation makes it difficult to identify emerging patterns, forecast demand, optimize pricing, and develop effective business strategies. Real-time analytics are critical for enabling agile and informed decision-making at the management level.

Sixth, AWMC experiences inefficiencies in order queuing, particularly during high-demand periods. The absence of a structured queuing system leads to disorganized order processing, causing confusion among staff and customers alike. Customers may face unclear order statuses, extended wait times, and dissatisfaction during pickup or delivery operations. An inefficient queuing process not only degrades the customer experience but also reduces overall business productivity by making it difficult to prioritize and sequence orders effectively.

Taken together, these operational challenges underscore the urgent need for a comprehensive web-based ordering and management system that can address AWMC's current inefficiencies and position the company for sustainable growth in the digital economy.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design, develop, and evaluate a web-based online ordering system with integrated membership management for Amazing World Marketing Corporation (AWMC). The proposed system aims to automate core business operations including order processing, inventory management, membership administration, payment handling, queue management, and sales analytics, thereby enhancing operational efficiency, reducing human error, and improving the overall experience of both customers and administrators.

1.3.2 Specific Objectives

In pursuit of the general objective, this study specifically aims to:

1. Conduct a comprehensive analysis of AWMC's current operational workflows, technical infrastructure, and user requirements in order to define precise system specifications. This involves documenting existing processes related to order management, inventory tracking, payment handling, and customer communication, as well as gathering feedback from key stakeholders including management, staff, and customers.
2. Design and develop a responsive, user-friendly web interface with separate, role-appropriate dashboards for administrators and customers. The customer interface will enable product browsing, cart management, order placement, and account management, while the administrator interface will provide tools for managing products, orders, user accounts, inventory, and system settings. The interface will be developed using HTML5, CSS3, and JavaScript to ensure accessibility and responsiveness across multiple device types.

3. Implement a reliable and scalable backend system using PHP and MySQL to support core business logic, including user authentication and authorization, order lifecycle management, inventory tracking, membership validation, and payment processing. The backend will be designed with security best practices including protection against SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF) attacks, as well as session management and input validation.
4. Develop an automated queue numbering system to improve the efficiency and transparency of order processing. This feature will assign unique queue numbers to customer orders, enabling both staff and customers to monitor the status of each transaction in real time, thereby reducing confusion, minimizing wait times, and enhancing the overall customer experience.
5. Integrate multiple payment gateways including GCash, PayMaya, PayPal, cash on pickup, and cash on delivery to provide customers with a diverse range of payment options. By accommodating both digital and traditional payment preferences, the system will cater to a broader customer base and improve transaction completion rates.
6. Develop a real-time administrative analytics dashboard that displays key performance indicators including total revenue, order volume, best-selling products, membership conversion rates, and payment method distribution. This dashboard will enable management to monitor business performance, identify trends, and make data-driven decisions in a timely and informed manner.
7. Conduct comprehensive system testing including unit testing, integration testing, security testing, performance testing, and user acceptance testing (UAT) to verify that

the system meets functional requirements, is secure and reliable, and delivers a satisfactory user experience for both administrators and customers.

1.4 Scope and Delimitation

1.4.1 Scope

This study encompasses the design, development, testing, and evaluation of a web-based ordering system with integrated membership management capabilities, tailored specifically to the operational requirements of Amazing World Marketing Corporation (AWMC). The system will support the following core functional areas: online product browsing and ordering, shopping cart management, automated order processing and queue numbering, multi-gateway payment integration, real-time inventory tracking, tiered membership management, and an administrative analytics and reporting dashboard.

The system will serve two primary user groups: customers (including both regular and member accounts) and administrators. Customers will be able to register for accounts, browse product catalogs, add items to their shopping carts, select payment methods, track order statuses, and manage their membership profiles. Administrators will have access to tools for managing products, processing and updating order statuses, managing customer accounts and membership tiers, monitoring inventory levels, and reviewing sales analytics.

The system will be developed using PHP for server-side logic, MySQL for database management, and HTML5, CSS3, and JavaScript for the front-end interface. It will be designed for deployment within the Philippine context, specifically for AWMC's staff and customer base. All development and initial testing will be conducted using XAMPP as the local development environment, with Visual Studio Code as the primary code editor.

1.4.2 Delimitations

The scope of this study is subject to several delimitations. First, the system will not integrate with external logistics or third-party shipping providers. Order delivery logistics will remain the responsibility of AWMC's existing fulfillment processes. Second, the system will not incorporate advanced artificial intelligence or machine learning features, such as automated product recommendation engines or predictive demand forecasting algorithms, as these fall outside the scope of the current development phase.

Third, while the system will integrate multiple payment gateways, actual payment processing will be implemented in a simulated or sandbox environment for the purposes of this study, and full production integration with payment service providers will be subject to AWMC's commercial agreements with those providers. Fourth, the system is designed for use by AWMC's existing customer base and does not include a public-facing marketplace or platform for external third-party sellers. Fifth, mobile application development for iOS or Android is not within the scope of this project; the system will be web-based and optimized for use through standard web browsers on desktop and mobile devices.

1.5 Significance of the Study

The development and implementation of the AWMC Ordering System is expected to generate significant benefits for multiple stakeholders, including the company's management and staff, its customers and members, system administrators, and the broader academic community. The following discussion outlines the anticipated contributions of this study to each stakeholder group.

1.5.1 AWMC Management and Staff

For AWMC's management team, the proposed system will provide real-time visibility into sales performance, inventory levels, and customer activity through an integrated analytics dashboard. This will enable managers to make data-driven decisions regarding product procurement, pricing strategies, membership incentives, and business planning. The automation of order processing and inventory tracking will significantly reduce the manual workload currently borne by staff, freeing personnel to focus on higher-value activities such as customer relationship management and strategic development. The reduction of human error in order management will also improve the accuracy of financial records, supporting more reliable accounting and audit processes.

1.5.2 Customers and Members

For AWMC's customers, the proposed system will deliver a substantially improved shopping experience. Customers will be able to browse the full product catalog, place orders, and complete transactions at any time and from any location, eliminating the need for in-person visits. The integration of multiple digital payment options including GCash, PayMaya, and PayPal will provide flexibility to accommodate a range of payment preferences. Real-time order tracking features will keep customers informed of their order status, reducing uncertainty and improving satisfaction. Membership functionality will provide registered members with access to exclusive products, tiered discounts, and personalized promotions, strengthening customer loyalty and encouraging repeat business.

1.5.3 System Administrators

For system administrators, the proposed platform will provide a centralized, organized interface for managing all aspects of AWMC's digital operations. Through an intuitive administrative dashboard, administrators will be able to manage product listings, monitor order queues, update inventory, administer customer accounts, and review performance metrics in

real time. The system's role-based access control will ensure that users interact only with the modules and data relevant to their responsibilities, reducing the risk of unauthorized access or accidental modification of critical data. Automated processes and structured data management will also reduce the risk of errors and data loss associated with manual record-keeping.

1.5.4 Future Researchers and Students

This study contributes to the academic literature on web-based business systems, e-commerce, and digital transformation in the Philippine SME context. It demonstrates how established principles from software engineering, management information systems, and e-commerce research can be applied to develop practical solutions for real-world business problems. Students and researchers in the fields of information technology and computer science may draw on this study as a reference for best practices in system design, database architecture, and full-stack web development. Additionally, the study may serve as a foundation for future research on topics such as AI-powered recommendation systems, mobile application development for direct-selling businesses, and the impact of digital ordering systems on distributor retention and customer loyalty.

1.6 Theoretical Framework

The design and development of the AWMC Ordering System is grounded in three foundational theoretical frameworks drawn from the fields of e-commerce, management information systems, and software engineering. These frameworks provide the conceptual basis for the system's core design choices and functional requirements.

1.6.1 Theory of Planned Behavior (Ajzen, 1991)

The Theory of Planned Behavior (TPB), originally proposed by Icek Ajzen, posits that an individual's intention to perform a behavior is determined by three factors: attitude toward the behavior, subjective norms, and perceived behavioral control. In the context of e-commerce adoption, Pavlou and Fygenson (2006) extended the TPB to explain how users decide whether to engage with online shopping platforms. Their research identified trust in payment security and the perceived usefulness and ease of use of the platform as critical determinants of purchase intention and completion.

This theoretical insight directly informed the design of the AWMC Ordering System's payment module and user interface. By integrating trusted and widely adopted payment platforms such as GCash, PayMaya, and PayPal, the system seeks to build user confidence and reduce perceived risk in online transactions. The emphasis on intuitive navigation, clear product information, and transparent order tracking is likewise intended to maximize perceived ease of use and utility, thereby reducing barriers to adoption and encouraging regular engagement with the platform.

1.6.2 Management Information Systems Theory (Laudon & Laudon, 2020)

Management Information Systems (MIS) theory, as articulated by Kenneth C. Laudon and Jane P. Laudon, holds that organizations can achieve significant competitive advantages by implementing information systems that centralize data, automate routine processes, and support informed decision-making at all levels of management. Laudon and Laudon emphasize that effective MIS implementations reduce redundancy, improve coordination across business functions, and provide managers with timely and accurate information for strategic planning.

The AWMC Ordering System operationalizes these principles through its integrated analytics dashboard, centralized database architecture, and automated order and inventory

management modules. By consolidating all transaction, inventory, and membership data in a single relational database, the system eliminates the fragmentation and redundancy inherent in AWMC's current manual processes. The real-time reporting capabilities of the analytics dashboard align directly with Laudon and Laudon's emphasis on timely information as a driver of organizational performance.

1.6.3 Software Engineering Principles (Pressman & Maxim, 2020)

The Software Engineering principles articulated by Roger S. Pressman and Bruce R. Maxim provide the methodological foundation for the design and development of the AWMC Ordering System. Pressman and Maxim advocate for structured development methodologies characterized by modular decomposition, iterative refinement, and adherence to established design patterns. These principles ensure that complex software systems are developed in a systematic, maintainable, and scalable manner.

In alignment with these principles, the AWMC Ordering System employs a three-layer architecture comprising a Presentation Layer (HTML5, CSS3, JavaScript), an Application Layer (PHP), and a Data Layer (MySQL). This separation of concerns promotes maintainability and scalability, as each layer can be updated or extended independently without disrupting the others. The Agile development methodology adopted for this project further reflects Pressman and Maxim's emphasis on iterative development, continuous testing, and responsiveness to changing requirements.

1.7 Conceptual Framework

The conceptual framework of this study illustrates the relationships between the system's inputs, processes, and outputs, following the Input-Process-Output (IPO) model commonly applied in systems development research.

Inputs

The inputs to the AWMC Ordering System include: user registration and authentication data submitted by customers and administrators; product information including product names, descriptions, prices, images, and stock quantities entered by administrators; customer orders including selected products, quantities, delivery preferences, and chosen payment methods; payment transaction data from integrated gateways; and administrative actions including product updates, order status changes, and membership adjustments.

Processes

The system's processing functions include: user authentication and role-based access control; product catalog management and inventory tracking; shopping cart management and order processing; automated queue number generation; payment processing and transaction recording; membership tier validation and management; and real-time analytics computation and dashboard rendering. These processes are governed by the business logic implemented in the PHP application layer and the relational data model maintained in the MySQL database.

Outputs

The outputs generated by the system include: confirmed order records with unique queue numbers and real-time status updates accessible to both customers and administrators; updated inventory levels automatically adjusted upon order completion; membership status updates reflecting tier transitions and earned benefits; processed payment records for each completed transaction; and analytical reports and dashboard visualizations displaying key performance indicators including revenue trends, order volumes, best-selling products, and

membership conversion rates. Together, these outputs enable AWMC to operate with greater efficiency, transparency, and data-driven decision-making capability.

1.8 Definition of Terms

The following terms are defined operationally as they are used in the context of this study:

AWMC Ordering System. A web-based platform developed in this study that enables customers to browse products, place orders, and complete transactions through multiple payment channels. The system includes integrated features for queue management, real-time inventory tracking, membership administration, and sales analytics, specifically designed to support the operational requirements of Amazing World Marketing Corporation.

Member. A registered user of the AWMC Ordering System who has created an account and is enrolled in the company's tiered membership program. Members are categorized into five tiers—Silver, Gold, Ruby, Emerald, and Diamond—each conferring varying levels of benefits including exclusive product access, discounted pricing, and promotional offers.

Queue Number. An automatically generated unique identifier assigned to each customer order upon placement. Queue numbers are used to organize and manage the sequence of order processing, providing both customers and staff with clear visibility into transaction status during pickup or delivery operations.

Middleware. PHP-based components within the system that function as an intermediary between the front-end user interface and the back-end database. Middleware handles critical

processes including user authentication, session management, authorization, input validation, and protection against common web security vulnerabilities.

API (Application Programming Interface). A set of PHP-based interfaces that facilitate communication between different components of the AWMC Ordering System. The API manages operations including cart item management, order processing, transaction status updates, and real-time data delivery to the administrative dashboard.

Database Gateway. The MySQL-based data management layer responsible for storing, retrieving, and organizing all persistent system data, including user accounts, product information, orders, transactions, and membership records. The database gateway ensures data integrity, consistency, and efficient accessibility across all system operations.

Ardeur de France. A proprietary fragrance product line offered by AWMC, inspired by internationally recognized perfume brands. This product line is available for both male and female consumers and is featured among the orderable items within the AWMC Ordering System.

GCash / PayMaya / PayPal. Digital payment platforms integrated into the AWMC Ordering System to facilitate secure and convenient online transactions. These platforms are widely adopted in the Philippines and allow customers to complete purchases electronically, providing flexibility for users who prefer cashless payment methods.

Cash on Pickup (COP). A payment method that allows customers to settle their order payment in cash at the time of collecting their purchase from AWMC's designated pickup location. This option accommodates customers who prefer traditional cash-based transactions.

Cash on Delivery (COD). A payment option in which customers pay for their orders in cash upon receiving delivery at their specified address. This method provides an accessible alternative for customers who do not have access to digital payment platforms.

Sales Analytics. The process of collecting, processing, and presenting sales data within the administrative dashboard of the AWMC Ordering System. Analytics outputs include metrics such as total revenue, transaction counts, best-selling products, membership conversion rates, and payment method distributions, which are used to support managerial decision-making.

User Interface (UI). The visual component of the AWMC Ordering System through which users interact with the platform. The UI encompasses menus, buttons, forms, dashboards, and product displays, designed to deliver an intuitive and seamless experience for both customers and administrators.

Authentication. The process of verifying the identity of users prior to granting access to the system. Authentication in the AWMC Ordering System relies on unique user credentials including email addresses and hashed passwords, supplemented by session management controls.

Authorization. The process of determining and enforcing the level of access granted to authenticated users based on their assigned roles. The AWMC Ordering System implements role-based access control to ensure that customers and administrators are permitted to interact only with the modules and data relevant to their respective responsibilities.

CHAPTER 2 REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of related foreign and local literature and studies relevant to the design, development, and evaluation of the AWMC Ordering System. The review is organized thematically to examine foundational concepts from e-commerce theory, management information systems, software engineering, loyalty program research, and Philippine digital transformation policy. The insights drawn from these sources provide the theoretical and empirical grounding for the system's design choices and anticipated contributions.

2.1 Foreign Literature

2.1.1 Electronic Commerce and Digital Ordering Systems

The transformation of business operations through digital technology has been extensively documented in the academic literature on electronic commerce. Turban et al. (2018), in their seminal work *Electronic Commerce 2018: A Managerial and Social Networks Perspective*, argue that digital ordering systems fundamentally enhance cost-efficiency and operational effectiveness by automating transactional workflows and reducing the overhead associated with manual processing. The authors contend that e-commerce platforms extend beyond the mere facilitation of transactions to encompass communication, information exchange, and relationship management between businesses and their customers.

Turban et al. further emphasize that customer satisfaction in digital ordering environments is closely linked to the speed and accuracy of order fulfillment, the reliability of the platform, and the clarity of communication regarding order status. Enhanced communication channels, real-time order tracking, and streamlined transaction processes

collectively contribute to a superior user experience that builds customer trust and encourages repeat patronage. These principles are directly reflected in the design of the AWMC Ordering System, which prioritizes transparency, accuracy, and ease of use across all customer-facing functions.

The literature on digital commerce also highlights the importance of multi-channel integration in modern ordering systems. As consumer behavior increasingly spans both online and offline channels, businesses must design ordering platforms capable of accommodating diverse customer preferences for product discovery, ordering, and payment. The AWMC Ordering System addresses this requirement by supporting multiple payment modalities and providing both online and pickup fulfillment options, ensuring accessibility for customers with varying preferences and levels of digital literacy.

2.1.2 Management Information Systems and Decision Support

Kenneth C. Laudon and Jane P. Laudon, in their authoritative text *Management Information Systems: Managing the Digital Firm* (2020), establish that robust management information systems (MIS) are essential enablers of informed business decision-making. They argue that organizations which successfully implement integrated information systems gain the ability to analyze operational data in real time, identify performance trends, and respond proactively to emerging challenges. The centralization of data across business functions eliminates redundancy, improves coordination, and provides managers with a unified and accurate picture of organizational performance.

Laudon and Laudon's framework for MIS design emphasizes the importance of aligning system capabilities with strategic business objectives. In the context of AWMC, this alignment is achieved through the integration of an analytics dashboard that provides

management with immediate access to key performance metrics including revenue, order volume, inventory status, and membership conversion rates. By transforming raw transactional data into actionable intelligence, the system enables AWMC's management team to make well-informed decisions regarding product procurement, promotional strategies, and distributor incentive programs.

O'Brien and Marakas (2011), in their complementary work *Management Information Systems*, further highlight the critical role of automated inventory management systems in maintaining operational effectiveness. They argue that real-time stock tracking prevents the twin inefficiencies of overstocking, which ties up capital and storage resources, and stockouts, which result in lost sales and damaged customer relationships. By automating inventory updates in response to order placements and administrative adjustments, the AWMC Ordering System addresses both of these concerns and provides management with the accurate stock data needed to optimize procurement decisions.

2.1.3 Software Engineering Principles and Web Application Development

Roger S. Pressman and Bruce R. Maxim, in *Software Engineering: A Practitioner's Approach* (2020), advocate for the application of structured software engineering methodologies in the development of web-based applications. Key principles advocated by Pressman and Maxim include modular decomposition, which involves breaking complex systems into discrete, independently testable components; iterative refinement, which involves progressively improving system functionality through repeated cycles of development and testing; and the application of established design patterns, which provide proven templates for addressing common software engineering challenges.

The application of these principles in the development of the AWMC Ordering System is manifested in its three-layer architectural design, which separates the presentation, application, and data layers to facilitate independent development, testing, and maintenance of each component. The adoption of the Agile software development methodology further reflects Pressman and Maxim's emphasis on iterative development, enabling the development team to respond to evolving requirements and to identify and resolve issues at an early stage of the development cycle.

The literature on web application security, closely related to software engineering practice, also informs the design of the AWMC Ordering System. Widely recognized security vulnerabilities in web applications, including SQL injection, cross-site scripting, and cross-site request forgery, are explicitly addressed in the system's application layer through input validation, parameterized queries, and session management controls. These security measures align with the principle that software quality encompasses not only functional correctness but also reliability, security, and resistance to malicious exploitation.

2.2 Foreign Studies

2.2.1 User Adoption of E-Commerce Platforms

Pavlou and Fygenson (2006), in their influential study published in *MIS Quarterly*, extended Ajzen's Theory of Planned Behavior to examine the factors that determine user adoption and continued use of e-commerce platforms. The authors identify trust in payment security and the perceived reliability of the platform as the most critical predictors of purchase completion among online shoppers. In addition, perceived ease of use and perceived usefulness are identified as significant antecedents of purchase intention, suggesting that platforms which

minimize navigational complexity and clearly communicate their value to users are more likely to achieve high adoption rates.

These findings have direct implications for the design of the AWMC Ordering System. The integration of trusted and widely recognized payment platforms—including GCash, PayMaya, and PayPal—is intended to address the trust dimension identified by Pavlou and Fygenon, reducing the perceived risk associated with online transactions and thereby encouraging purchase completion. The emphasis on intuitive interface design, clear product information presentation, and transparent order tracking further addresses the usability dimension, ensuring that the system is accessible and comprehensible to users regardless of their prior experience with digital commerce platforms.

2.2.2 Loyalty Programs and Tiered Membership Structures

Empirical research on customer loyalty programs consistently demonstrates that tiered reward structures are among the most effective mechanisms for increasing customer retention and encouraging repeat purchasing behavior. Studies in this domain show that tiered programs, which offer escalating benefits as customers advance through successive levels of membership, create powerful incentives for sustained engagement. The anticipation of achieving a higher membership tier, with its associated benefits, motivates customers to maintain or increase their purchasing activity over time, generating a compounding effect on customer lifetime value.

In the context of direct-selling businesses, tiered membership structures serve a dual function: they simultaneously incentivize end-customer purchasing and motivate distributor performance. By linking membership advancement to cumulative sales volume or referral activity, companies can align the interests of individual distributors with the organization's overall growth objectives. The AWMC Ordering System's five-tier membership model—

comprising Silver, Gold, Ruby, Emerald, and Diamond levels—is designed to operationalize these principles by providing AWMC’s distributors and customers with clear, progressive incentives for long-term loyalty and engagement.

Research by Kumar and Shah (2004) further establishes that loyalty program participants exhibit higher average transaction values, greater purchase frequency, and lower churn rates compared to non-participants. These behavioral differences generate measurable economic benefits for the business in the form of increased revenue per customer and reduced customer acquisition costs. The AWMC Ordering System’s membership management module is designed to capture and analyze these behavioral patterns, providing management with data-driven insights into the effectiveness of the company’s loyalty initiatives.

2.3 Local Literature

2.3.1 Philippine Digital Transformation Policy

The Department of Information and Communications Technology (DICT), established under Republic Act No. 10844, is the primary government agency responsible for formulating and implementing national policies, plans, and programs related to information and communications technology in the Philippines. The DICT leads the Philippine Digital Transformation Strategy 2022, a comprehensive national initiative designed to leverage ICT to modernize government services, strengthen the economy, and improve the quality of life of Filipino citizens, with particular emphasis on enabling small and medium enterprises to adopt and benefit from digital technologies.

The DICT’s mandate encompasses a vast constituency: 82 provinces, approximately 1,700 local government units, around 115 million citizens, approximately 2 million SMEs, and over 42,000 barangays across the archipelago. This scale underscores the critical importance

of promoting digital literacy and technology adoption at all levels of Philippine society, and highlights the significant potential impact of digital transformation initiatives on the country's economic development. For companies like AWMC, the Philippine Digital Transformation Strategy 2022 represents both a policy mandate and an enabling framework that supports the adoption of digital ordering and management systems.

2.3.2 The Philippine Direct-Selling Industry

The direct-selling industry occupies a significant position in the Philippine economy, characterized by a large and distributed network of independent distributors who market and sell products directly to consumers. Data from Agriculture and Agri-Food Canada indicate that the retail value of direct selling in the Philippines reached approximately 1.37 billion US dollars in 2021, with projections indicating continued growth to approximately 1.75 billion US dollars by 2026. These figures reflect both the established importance of the direct-selling sector and its strong growth trajectory, driven by increasing consumer demand for health, wellness, and personal care products.

The Direct Selling Association of the Philippines (DSAP) further documents the industry's role as a significant source of entrepreneurial opportunity and supplementary income for hundreds of thousands of Filipinos. The MLM and direct-selling model, which AWMC employs, relies on structured distributor networks in which members are motivated not only by product commissions but also by the incentives associated with building and managing their own downline organizations. Effective management of these distributor networks, including the administration of tiered membership benefits, compensation plans, and performance tracking, is therefore a central operational challenge for companies operating in this sector.

2.4 Local Studies

2.4.1 Digital Adoption among Philippine SMEs

Medium-sized enterprises in the Philippines account for approximately 99.5 percent of registered businesses, generate about 65 percent of total employment, and contribute around 25 percent of export revenues, according to the Joint SDG Fund (2024). Despite their overwhelming economic significance, many of these enterprises continue to experience low levels of digital adoption and face persistent productivity challenges rooted in manual, labor-intensive business processes. Research consistently identifies the adoption of cloud-based platforms, digital payment systems, and automated inventory management tools as key levers for improving productivity, reducing costs, and enabling Philippine SMEs to compete more effectively in the digital economy.

Barriers to digital adoption among Philippine SMEs include limited financial resources for technology investment, inadequate digital literacy among management and staff, concerns about data security, and the perceived complexity of implementing and maintaining digital systems. These barriers highlight the importance of designing digital solutions that are not only functionally robust but also intuitive, cost-effective, and well-supported by appropriate training and documentation. The AWMC Ordering System addresses these considerations through its emphasis on user-friendly interface design, clear documentation, and compatibility with commonly available hardware and software infrastructure.

2.4.2 Distributor Retention in Philippine Direct-Selling Companies

Distributor retention represents one of the most persistent operational challenges facing direct-selling companies in the Philippines. Research cited by Triple i Consulting (2025), drawing on data from Euromonitor International, indicates that distributors frequently disengage from direct-selling organizations due to insufficient income, inadequate training and

support, and a perceived lack of recognition for their contributions. These findings underscore the importance of structured incentive programs, transparent performance tracking, and meaningful reward systems in maintaining distributor motivation and commitment.

Tiered membership programs are identified in the literature as one of the most effective mechanisms for addressing distributor retention challenges. By creating a clear and attainable pathway from entry-level membership to higher-tier status, companies can provide distributors with ongoing motivation to increase their activity and build their networks. The visibility of membership tier benefits and the recognition associated with advancing to higher levels also contribute to a sense of achievement and belonging that reinforces distributor loyalty. The AWMC Ordering System's membership management module is specifically designed to support these retention mechanisms by providing transparent, automated tracking of member activity and tier status.

2.5 Synthesis of the Reviewed Literature and Studies

The reviewed foreign and local literature and studies converge on several key insights that directly inform the design and development of the AWMC Ordering System. First, the literature consistently establishes that digital ordering systems provide significant operational benefits for businesses that depend on complex distributor networks or membership structures, including reductions in processing time, error rates, and administrative overhead. Second, user adoption of e-commerce platforms is critically dependent on trust in payment security, perceived ease of use, and perceived usefulness—factors that have been explicitly addressed in the system's design through the integration of trusted payment gateways and an emphasis on intuitive interface design.

Third, the literature on loyalty programs provides strong empirical support for the effectiveness of tiered membership structures in fostering customer retention and distributor engagement, validating the decision to make membership management a central feature of the AWMC Ordering System. Fourth, Philippine-specific research and policy documents highlight the urgency and opportunity of digital transformation for SMEs operating in the direct-selling sector, situating the AWMC Ordering System within a broader national context of technological modernization and economic development.

Fifth, software engineering principles advocated by Pressman and Maxim provide a methodological framework for ensuring that the system is developed in a systematic, scalable, and maintainable manner, with quality assurance processes integrated throughout the development lifecycle. Finally, the MIS theory of Laudon and Laudon establishes the strategic importance of real-time data access and integrated information systems for effective organizational decision-making, which is operationalized in the system's analytics dashboard and centralized database architecture.

Taken together, these sources define the functional and non-functional requirements that the AWMC Ordering System must satisfy, and provide a robust theoretical and empirical foundation for the system's design choices. The system is accordingly positioned as a practical, theoretically grounded solution that addresses AWMC's identified operational challenges while contributing to the broader goal of digital transformation in the Philippine direct-selling industry.

CHAPTER 3 METHODOLOGY

This chapter presents a detailed description of the research design, system development methodology, system architecture, features and modules, database design, development tools and technologies, data gathering techniques, testing procedures, and evaluation criteria employed in the development of the AWMC Ordering System. The methodology is designed to ensure that the system is developed in a rigorous, systematic, and user-centered manner that meets the functional requirements identified in Chapter 1 and is grounded in the theoretical and empirical foundations established in Chapter 2.

3.1 Research Design

This study employs a developmental or systems development research design, which is the most appropriate approach for studies that involve the design, construction, and evaluation of a functional artifact intended to address real-world operational problems. Developmental research designs are characterized by a systematic cycle of requirements analysis, system design, implementation, testing, and evaluation, with iterative refinement based on testing outcomes and stakeholder feedback. This design is well-suited to the objectives of the current study, which centers on the creation and empirical evaluation of a web-based ordering and management system for AWMC.

The developmental research paradigm is complemented by an applied research orientation, in which the practical utility and real-world applicability of the developed system are primary concerns. The system is evaluated not only in terms of its technical correctness and functional completeness but also with respect to its usability, security, performance, and reliability from the perspective of its intended users. This dual focus on technical quality and

practical utility ensures that the research outcomes are directly applicable to AWMC's operational needs and contribute meaningfully to the organization's digital transformation objectives.

3.2 System Development Methodology

The AWMC Ordering System will be developed using the Agile Software Development methodology. Agile is an iterative, incremental approach to software development that emphasizes collaboration, flexibility, and continuous improvement. Unlike traditional waterfall methodologies, which require all requirements to be fully specified before development begins, Agile allows the development team to respond to evolving requirements and stakeholder feedback throughout the development process. This adaptability is particularly valuable in the context of the current project, where user requirements may be refined as stakeholders gain familiarity with the proposed system's capabilities.

The development process will be organized into five sequential phases, each corresponding to a distinct stage in the system's lifecycle:

Phase 1: Requirements Analysis

The requirements analysis phase involves the systematic collection and documentation of all functional and non-functional requirements for the AWMC Ordering System. This phase employs multiple data collection techniques, including structured interviews with AWMC management and staff, direct observation of existing ordering and inventory management workflows, document analysis of current order records and product catalogs, and surveys administered to current members and customers. The outputs of this phase include a comprehensive list of user stories, functional requirements, non-functional requirements, and technical specifications that serve as the foundation for all subsequent development activities.

Phase 2: System Design

The system design phase translates the requirements identified in Phase 1 into a detailed blueprint for the proposed system. This phase includes the design of the overall system architecture, the user interface layout and navigation structure, the relational database schema, and the integration points between the system's various modules. Design artifacts produced during this phase include wireframes and prototypes of the user interface, entity-relationship diagrams (ERDs) illustrating the database structure and entity relationships, and data flow diagrams (DFDs) depicting how data moves through the system. These artifacts are reviewed and validated with key stakeholders before development begins to ensure alignment with user expectations and business requirements.

Phase 3: Development

The development phase involves the iterative implementation of the system's front-end and back-end components in short development sprints. Each sprint focuses on the implementation and initial testing of a specific system module or feature, such as user authentication, product catalog management, shopping cart functionality, order processing, payment integration, or membership management. Regular code reviews are conducted to ensure adherence to coding standards and to identify potential issues at an early stage. Incremental testing is performed throughout the development phase to verify that each implemented component functions correctly and integrates smoothly with the rest of the system.

Phase 4: Testing

The testing phase involves comprehensive evaluation of the system across multiple testing dimensions to verify its functional correctness, security, performance, and usability. Testing activities include unit testing of individual PHP functions and database queries,

integration testing of the interactions between system layers and modules, security testing for common web application vulnerabilities, performance testing under simulated multi-user conditions, and user acceptance testing (UAT) with actual AWMC administrators and customers. Defects identified during testing are documented, prioritized, and resolved through targeted fixes and retesting cycles. The outputs of the testing phase include detailed test reports documenting the results of each testing activity and the actions taken to address identified defects.

Phase 5: Deployment (Prototype)

The deployment phase involves the implementation of a functional prototype of the AWMC Ordering System in a controlled environment for demonstration and evaluation by stakeholders. This prototype showcases the system's core features and allows users to interact with the system in a realistic operational context. Feedback gathered during the prototype demonstration is used to identify areas for improvement and to guide further refinement of the system prior to full production deployment. The deployment phase serves as a critical validation step that ensures the system meets user expectations and satisfies AWMC's operational requirements before it is released for live use.

3.3 System Architecture

The AWMC Ordering System is built upon a three-layer architecture that ensures a clear separation of concerns, supports maintainability, and provides a scalable foundation for future system enhancements. The three layers—Presentation, Application, and Data—are described in detail below.

3.3.1 Presentation Layer

The Presentation Layer constitutes the front end of the AWMC Ordering System and is implemented using HTML5, CSS3, and JavaScript. This layer is responsible for rendering the user interface and facilitating all direct interactions between the user and the system. For customers, the Presentation Layer provides interfaces for product browsing, shopping cart management, order placement, payment selection, order history review, and profile management. For administrators, it renders the administrative dashboard, product management interfaces, order management tools, user account management screens, and the analytics and reporting dashboard.

The Presentation Layer is designed with responsiveness as a core principle, ensuring that the interface adapts appropriately to a range of screen sizes and device types, including desktop computers, tablets, and smartphones. This responsiveness is essential for accommodating the diverse technological contexts in which AWMC's customers and staff access the system. Accessibility considerations, including clear labeling of interactive elements, consistent navigation patterns, and appropriate color contrast, are incorporated into the interface design to minimize the learning curve for new users and to support users with varying levels of digital literacy.

3.3.2 Application Layer

The Application Layer functions as the back end of the AWMC Ordering System and is implemented using PHP. This layer is responsible for all core business logic processing, including user authentication and session management, order lifecycle management, stock availability verification, queue number generation, membership tier validation and management, and payment processing coordination. The Application Layer serves as the intermediary between the Presentation Layer and the Data Layer, receiving inputs from the

user interface, processing them according to the system's business rules, and returning appropriate outputs to be rendered by the front end.

Security is a primary concern in the design of the Application Layer. Protective measures implemented at this layer include parameterized database queries to prevent SQL injection attacks, output encoding to mitigate cross-site scripting (XSS) vulnerabilities, CSRF token validation to protect against cross-site request forgery, session timeout controls to prevent unauthorized access through abandoned sessions, rate limiting on login attempts to resist brute-force attacks, and role-based access control to ensure that each user can only access the modules and data corresponding to their assigned role.

3.3.3 Data Layer

The Data Layer is implemented using MySQL and manages all persistent data storage and retrieval operations within the AWMc Ordering System. The MySQL database is designed in accordance with relational database normalization principles to minimize data redundancy, ensure data integrity, and facilitate efficient querying. Referential integrity constraints are enforced through the use of foreign key relationships between related tables, ensuring that the database remains internally consistent even as data is created, updated, and deleted through normal system operations.

The database is also designed with performance optimization in mind. Appropriate indexing is applied to frequently queried columns to accelerate data retrieval operations, which is particularly important for high-traffic operations such as product catalog browsing, order status queries, and analytics dashboard rendering. The relational data model supports complex queries that span multiple tables, enabling the analytics module to generate comprehensive performance reports from a single, unified data source.

3.4 System Features and Modules

The AWMC Ordering System comprises seven integrated core modules, each responsible for a distinct functional domain within the system. These modules are designed to work together seamlessly to deliver a comprehensive ordering and management platform.

3.4.1 Authentication Module

The Authentication Module manages all aspects of user identity verification and session management within the AWMC Ordering System. This module handles user registration (including the collection and validation of account information, and the secure hashing of passwords using bcrypt), user login (including credential verification and session initialization), and user logout (including session termination and token invalidation). The module also enforces role-based access control, ensuring that customers and administrators are directed to their respective interfaces and are prevented from accessing functionality outside the scope of their assigned role.

3.4.2 Product Management Module

The Product Management Module enables administrators to create, update, and delete product listings within the AWMC catalog. Products are organized into three categories: loose products (available to all customers), member-only products (accessible exclusively to registered members), and membership packages (bundles that confer membership tier status upon purchase). For each product, administrators can specify the product name, description, unit price, product category, stock quantity, and an associated product image. The module enforces real-time validation of input data to ensure that product records remain accurate and complete.

3.4.3 Shopping Cart Module

The Shopping Cart Module provides each registered user with a personalized, session-persistent shopping cart in which selected products are accumulated prior to checkout. The module performs real-time validation of product availability when items are added to the cart, preventing customers from placing orders for out-of-stock products. Customers can adjust item quantities or remove products from their carts at any point before checkout. The module also applies membership-tier-appropriate pricing to qualifying products, ensuring that members receive the discounts and benefits associated with their tier level.

3.4.4 Order Processing Module

The Order Processing Module manages the complete lifecycle of customer orders, from initial placement through final completion. When a customer submits an order, the module records all relevant order details—including customer information, selected products and quantities, delivery method, payment method, and total amount—in the database, and assigns a unique queue number to the order for tracking purposes. The module then initiates the appropriate payment processing workflow based on the customer's chosen payment method and updates inventory stock levels to reflect the ordered quantities. Administrators can update order status through the administrative dashboard, with status changes reflected in real time on the customer's order tracking interface.

3.4.5 Membership Management Module

The Membership Management Module administers AWMC's five-tier membership program, comprising Silver, Gold, Ruby, Emerald, and Diamond levels. The module tracks each member's current tier status and cumulative activity, controls access to member-exclusive products and pricing tiers based on membership level, and processes membership package purchases that confer or upgrade tier status. Administrators can manually adjust membership

status through the administrative interface, enabling the system to accommodate special cases or corrections that may arise in the normal course of business operations.

3.4.6 Analytics and Sales Reporting Module

The Analytics and Sales Reporting Module provides administrators with a comprehensive, real-time dashboard that aggregates and visualizes key business performance metrics. Dashboard components include total revenue and revenue trends over configurable time periods, order volume and order status distributions, best-selling products by volume and revenue, membership tier distribution and conversion rates, and payment method usage statistics. Data visualizations are generated using the Chart.js JavaScript library, providing clear and interactive graphical representations of performance data. The module is designed to support rapid interpretation of business intelligence, enabling management to identify trends, diagnose operational issues, and make informed strategic decisions.

3.4.7 User Management Module

The User Management Module enables administrators to view, search, filter, and manage customer account records within the AWMC Ordering System. Administrative functions include reviewing complete customer profiles, updating membership status, resetting account credentials, and deactivating or removing user accounts where necessary. The module also provides administrators with visibility into each customer's order history, enabling them to identify high-value customers and to tailor membership incentives accordingly. Access to the User Management Module is restricted to authorized administrators to protect the privacy and security of customer data.

3.5 Database Design

The AWMC Ordering System's relational database is implemented in MySQL and comprises five primary tables, each designed to capture a distinct category of system data. The database schema is normalized to the third normal form (3NF) to minimize data redundancy and ensure data integrity. The following subsections describe each primary table and its attributes.

3.5.1 Users Table

The Users table stores account information for all registered system users, encompassing both customers and administrators. Key attributes include: `user_id` (primary key, auto-incremented), `full_name`, `email` (unique), `hashed_password`, `contact_number`, `delivery_address`, `membership_status` (enumeration: none, silver, gold, ruby, emerald, diamond), `role` (enumeration: customer, admin), and `created_at` (timestamp). Sensitive fields including `email`, `contact_number`, and `membership_status` are encrypted using AES-256-CBC encryption to protect user privacy in the event of unauthorized database access.

3.5.2 Products Table

The Products table maintains the complete product catalog for the AWMC Ordering System. Key attributes include: `product_id` (primary key), `product_name`, `description`, `price` (decimal), `product_type` (enumeration: loose, member, package), `image_path` (file path to the product image stored on the server), `stock_quantity` (integer), and `added_at` (timestamp). The `product_type` attribute is used by the system to enforce access controls, ensuring that member-only products are visible only to authenticated members and that membership packages are handled through the appropriate enrollment workflow.

3.5.3 Orders Table

The Orders table records complete transaction data for all customer orders placed through the AWMC Ordering System. Key attributes include: `order_id` (primary key), `user_id`

(foreign key referencing Users), customer_name, address, contact_number, order_method (enumeration: pickup, delivery), payment_method (enumeration: gcash, paymaya, paypal, cash_on_pickup, cash_on_delivery), payment_status (enumeration: pending, paid, failed), order_status (enumeration: pending, processing, ready, completed, cancelled), queue_number (auto-generated unique identifier), total_amount (decimal), and order_date (timestamp). The queue_number attribute is automatically assigned by the Order Processing Module upon order placement and is used to organize and track orders throughout their processing lifecycle.

3.5.4 Order Items Table

The Order_Items table stores the individual product line items associated with each customer order, implementing a one-to-many relationship with the Orders table. Key attributes include: item_id (primary key), order_id (foreign key referencing Orders), product_id (foreign key referencing Products), quantity (integer), and unit_price (decimal, recording the price at the time of purchase to preserve historical accuracy regardless of subsequent price changes). This table enables the system to calculate order totals, generate detailed order summaries, and produce accurate sales analytics by product.

3.5.5 User Payment Accounts Table

The User_Payment_Accounts table stores customers' saved payment account details for GCash and PayMaya, facilitating faster checkout for repeat customers. Key attributes include: account_id (primary key), user_id (foreign key referencing Users), account_type (enumeration: gcash, paymaya), account_name, account_number (encrypted), and is_default (boolean, indicating the customer's preferred payment method). Account numbers are stored in encrypted form to protect sensitive financial information. Customers may save multiple payment accounts and designate one as their default for expedited checkout.

3.6 Development Tools and Technologies

The following tools and technologies have been selected for the development of the AWMC Ordering System based on their suitability for the project's technical requirements, their compatibility with the team's existing competencies, and their widespread adoption in the web development community:

- **PHP:** A widely used server-side scripting language selected for all back-end development and middleware implementation. PHP provides robust support for web application development, including database connectivity, session management, form processing, and file handling.
- **MySQL:** An open-source relational database management system selected for data storage and retrieval. MySQL provides a reliable, high-performance platform for managing the structured data generated by the AWMC Ordering System, with strong support for referential integrity constraints, indexing, and complex queries.
- **HTML5, CSS3, JavaScript:** Client-side technologies used to implement the system's user interface. HTML5 provides the structural markup of web pages; CSS3 is used for visual styling and responsive layout; and JavaScript enables dynamic client-side interactions and asynchronous data loading.
- **Chart.js:** A JavaScript library used to generate data visualizations in the administrative analytics dashboard. Chart.js provides a range of chart types including line charts, bar charts, pie charts, and doughnut charts, with support for interactive tooltips and responsive sizing.

- **XAMPP:** A free, open-source local development environment that integrates Apache (web server), PHP, and MySQL into a single installation package. XAMPP is used for all local development and testing activities prior to deployment.
- **Visual Studio Code:** A lightweight but powerful source code editor used as the primary development environment. Visual Studio Code provides syntax highlighting, code completion, integrated Git support, and a rich ecosystem of extensions for PHP and web development.
- **Git and GitHub:** A distributed version control system (Git) and collaborative code hosting platform (GitHub) used to manage the project's source code, track changes over time, facilitate team collaboration, and maintain a complete history of all development activities.

3.7 Data Gathering Techniques

The requirements analysis phase of the study employs the following data gathering techniques to ensure that the AWMC Ordering System is designed to accurately address the company's operational needs and user requirements:

3.7.1 Structured Interviews

Structured interviews will be conducted with AWMC management and frontline staff to document existing ordering workflows, identify pain points and inefficiencies in the current system, and define priority requirements for the proposed solution. Interview questions will be prepared in advance to ensure comprehensive coverage of all relevant operational domains, while allowing for follow-up questions to explore issues in greater depth.

3.7.2 Direct Observation

Direct observation of the existing manual ordering process will be conducted to identify inefficiencies, bottlenecks, and opportunities for improvement that may not be fully articulated through interview alone. Observation will focus on the complete order lifecycle, from initial customer inquiry through order placement, payment collection, fulfillment, and delivery or pickup.

3.7.3 Document Analysis

Existing business documents, including product catalogs, order forms, membership registration documents, pricing schedules, and sales records, will be reviewed and analyzed to ensure that the system's data model and business logic accurately reflect AWMC's operational reality. Document analysis will also help to identify categories of data that must be captured and managed by the proposed system.

3.7.4 Surveys

Surveys will be distributed to current AWMC members and customers to collect feedback on their preferences, pain points, and desired features for the proposed ordering system. Survey instruments will be designed to capture both quantitative ratings and qualitative comments, providing the development team with a rich and representative picture of user needs and expectations.

3.8 Testing Procedures

The AWMC Ordering System will undergo comprehensive testing across five testing dimensions to verify its functional correctness, security, performance, usability, and reliability. Testing activities are described in detail below.

3.8.1 Unit Testing

Unit testing will verify the correct operation of individual PHP functions, modules, and database queries in isolation from the rest of the system. Each unit will be tested with a range of valid, invalid, and boundary inputs to confirm that it produces the expected outputs and handles error conditions appropriately. Unit tests will be documented and archived to support regression testing following any modifications to the system's codebase.

3.8.2 Integration Testing

Integration testing will validate the interactions and data flows between the system's various components, including the front-end interface, PHP application layer, and MySQL database. Integration tests will verify that data entered through the user interface is correctly processed by the application layer, accurately stored in the database, and correctly retrieved and displayed upon subsequent requests. Particular attention will be given to the integration points between modules, such as the handoff between the Shopping Cart Module and the Order Processing Module, and between the Order Processing Module and the Inventory Management functions.

3.8.3 Security Testing

Security testing will evaluate the system's resilience against common web application vulnerabilities, including SQL injection, cross-site scripting (XSS), cross-site request forgery (CSRF), session hijacking, and unauthorized access attempts. Testing will be conducted using both manual testing techniques and automated scanning tools where appropriate. Identified vulnerabilities will be documented, prioritized by severity, and remediated through targeted code fixes before the system is released for user acceptance testing.

3.8.4 Performance Testing

Performance testing will assess the system's ability to maintain acceptable response times and stability under simulated concurrent user loads representative of AWMC's expected

peak traffic conditions. Performance tests will measure page load times, database query execution times, and system throughput under varying load conditions. Results will be used to identify performance bottlenecks and to guide optimization efforts targeting database indexing, query efficiency, and server configuration.

3.8.5 User Acceptance Testing (UAT)

User acceptance testing will involve representative AWMC administrators and customers interacting with the deployed prototype system to evaluate its usability, functional completeness, and alignment with operational requirements. UAT participants will be asked to complete a series of predefined task scenarios covering all core system functions, and will be asked to provide structured feedback on their experience using a standardized evaluation instrument. Findings from UAT will be used to guide final refinements to the system prior to full deployment.

3.9 Evaluation Criteria

The AWMC Ordering System will be systematically evaluated against the following criteria using a standardized assessment instrument administered to UAT participants:

3.9.1 Functionality

The system must accurately and completely execute all defined functional requirements without errors or omissions. Evaluation of functionality will assess whether each system module performs its intended operations correctly across a representative range of input conditions and user scenarios.

3.9.2 Usability

The system's interface must be intuitive, consistent, and accessible for both customers and administrators, minimizing the need for extensive training or prior technical knowledge.

Usability will be evaluated through UAT participant feedback on navigation clarity, interface consistency, error message helpfulness, and overall ease of use.

3.9.3 Security

The system must effectively protect user data from unauthorized access and safeguard against common web application vulnerabilities. Security will be evaluated through the results of dedicated security testing activities and through an assessment of the comprehensiveness and effectiveness of the system's implemented security controls.

3.9.4 Performance

The system must maintain fast response times and operational stability under realistic concurrent user loads. Performance will be evaluated through the results of load and stress testing, with reference to predefined acceptable response time thresholds for key user interactions including product catalog browsing, order placement, and dashboard rendering.

3.9.5 Reliability

The system must consistently process orders correctly, maintain data integrity across all operations, and recover gracefully from unexpected inputs or error conditions. Reliability will be evaluated through the combined results of unit testing, integration testing, and UAT, with particular attention to the system's behavior under edge cases and error conditions.

3.10 Data Flow Diagram (DFD — Level 0)

The Level 0 Data Flow Diagram (context diagram) of the AWMC Ordering System illustrates the high-level data flows between the system and its two primary external entities: Customers and Administrators.

Customer data flows into the system include: registration details and login credentials submitted during account creation and authentication; product browsing requests; order

placement data including selected products, quantities, delivery preferences, and payment method selection; and payment transaction data submitted through integrated payment gateways. Data flows from the system to customers include: product catalog content and search results; order confirmations with assigned queue numbers; real-time order status updates; and membership tier information and associated benefits.

Administrator data flows into the system include: product management actions (additions, updates, deletions); order status updates; user account management actions; and system configuration changes. Data flows from the system to administrators include: real-time analytics dashboard data; order queue information; inventory status reports; customer account information; and sales performance reports.

3.11 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) of the AWMC Ordering System defines the core database entities, their attributes, and the relationships between them. The primary entities and their relationships are described as follows:

The User entity (`user_id`, `full_name`, `email`, `hashed_password`, `contact_number`, `address`, `membership_status`, `role`, `created_at`) is related to the Order entity through a one-to-many relationship, as a single user may place multiple orders over time. The User entity is also related to the User_Payment_Account entity through a one-to-many relationship, as a single user may save multiple payment account records.

The Product entity (`product_id`, `product_name`, `description`, `price`, `product_type`, `image_path`, `stock_quantity`, `added_at`) is related to the Order_Item entity through a one-to-many relationship, as a single product may appear as a line item in multiple orders.

The Order entity (order_id, user_id, customer_name, address, contact_number, order_method, payment_method, payment_status, order_status, queue_number, total_amount, order_date) is related to the Order_Item entity through a one-to-many relationship, as a single order may contain multiple product line items.

These relational structures ensure data integrity through referential constraints, support efficient querying across related data, and provide a scalable foundation for the future extension of the AWMC Ordering System's data model.

3.12 Use Case Diagram

The AWMC Ordering System's use case model identifies two primary actor roles—Customer and Administrator—and defines the functional interactions available to each role. The use cases associated with each actor are described below.

Customer Use Cases

- Register for a new account and log in to the system using verified credentials.
- Browse the product catalog, with access to products filtered based on membership tier.
- Search for specific products by name or category.
- Add selected products to the shopping cart and adjust quantities.
- Proceed to checkout, specifying delivery preferences and selecting a payment method.
- Complete payment through an integrated payment gateway or select cash payment at pickup or delivery.
- View the assigned order queue number and monitor real-time order status.

- Review past order history and order details.
- Update personal profile information and saved payment account details.
- Purchase a membership package to enroll in or upgrade within the membership tier system.

Administrator Use Cases

- Log in to the administrative dashboard using administrator credentials.
- Add new products to the catalog, update existing product information, and remove discontinued products.
- View the order queue, review order details, and update order status throughout the processing lifecycle.
- Manage customer accounts, including reviewing profiles, updating membership status, and deactivating accounts.
- Monitor real-time inventory levels and adjust stock quantities as needed.
- Access the analytics dashboard to review key performance metrics and generate sales reports.
- Configure system settings and manage administrative user accounts.

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